

ICSR Processing & Case Narrative Portfolio

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This portfolio demonstrates competency in Individual Case Safety Report (ICSR) processing, MedDRA coding, adverse event assessment, and case narrative writing. Three cases of varying complexity are presented - covering a drug-drug interaction, a pediatric hepatotoxicity case, and a biologic-induced immune-mediated reaction - processed in accordance with ICH E2A guidelines and international pharmacovigilance reporting standards.

Reference Standards Applied: ICH E2A | ICH E6(R3) | MedDRA Hierarchy | WHO-UMC Causality Categories

CASE 1: Drug-Drug Interaction - Warfarin + Fluconazole

Source Case Description

A 68-year-old male patient with atrial fibrillation and a recent fungal infection was on stable Warfarin therapy with INR maintained at 2.5. His physician prescribed Fluconazole 200mg daily for 7 days. On day 5 of Fluconazole, he presented to the ER with gum bleeding and hematuria. INR was found to be 6.8. Warfarin was withheld for 2 days and Vitamin K was administered. INR normalized to 2.9 within 4 days. Warfarin was restarted at a reduced dose. No prior bleeding history. Reporter: treating physician.

Section 1 - Case Validity

Assessment: VALID - All four ICH E2A minimum criteria met:

- Identifiable patient: 68-year-old male
- Identifiable reporter: Treating physician (medically qualified)
- Suspect drug: Fluconazole (interacting with Warfarin)
- Adverse event: Bleeding events, supratherapeutic INR

Section 2 - Patient & Drug Information

Patient: 68-year-old male

Indication: Atrial fibrillation; fungal infection

Suspect Drug: Fluconazole 200mg daily oral

Concomitant Drug: Warfarin (stable, INR 2.5 at baseline)

Action Taken: Warfarin withheld; restarted at reduced dose. Fluconazole course completed.

Outcome: Recovered - INR normalized to 2.9 within 4 days

Section 3 - MedDRA Coding

AE: Gum bleeding

PT: Gingival bleeding, HLT: Bleeding tendencies, HLGT: Coagulation and bleeding disorders, SOC: Blood and lymphatic system disorders

AE: Blood in urine

PT: Haematuria, HLT: Urinary tract signs and symptoms, HLGT: Urinary tract disorders, SOC: Renal and urinary disorders

AE: High INR

PT: International normalised ratio increased, HLT: Coagulation and bleeding investigations, HLGT: Haematological investigations, SOC: Investigations

Section 4 - Seriousness Assessment

Assessment: SERIOUS

Criteria met: Medically significant event requiring intervention to prevent permanent harm - Vitamin K administered, Warfarin withheld. INR of 6.8 represents significant, life-threatening bleeding risk.

Section 5 - Causality Assessment

Assessment: PROBABLE (WHO-UMC)

- Clear temporal relationship - bleeding onset on day 5 of Fluconazole
- Positive dechallenge - INR normalized after Warfarin withheld and Vitamin K administered

- Known mechanism - Fluconazole is a strong CYP2C9 inhibitor, reducing Warfarin metabolism and potentiating anticoagulant effect
- No alternative causative factor identified; no prior bleeding history
- Warfarin restarted at reduced dose confirms drug interaction as primary cause

Section 6 - Expectedness Assessment

Assessment: EXPECTED (Labeled)

Fluconazole prescribing information lists potentiation of Warfarin anticoagulant effect and bleeding as known drug interactions. Event is consistent with reference safety information.

Section 7 - Case Narrative

This is a spontaneous report from a treating physician concerning a 68-year-old male patient with a known history of atrial fibrillation and a recent fungal infection, with no prior history of bleeding disorders.

The patient was on stable Warfarin therapy with INR maintained at 2.5. Fluconazole 200mg daily was initiated for treatment of fungal infection. On day 5 of Fluconazole therapy, the patient presented to the emergency room with gum bleeding and hematuria. INR on presentation was measured at 6.8, indicating supratherapeutic anticoagulation.

Warfarin was withheld for 2 days and Vitamin K was administered. INR normalized to 2.9 within 4 days. Warfarin was subsequently restarted at a reduced dose. Full clinical recovery was reported.

This case is assessed as serious based on the requirement for medical intervention to prevent further harm. Causality is assessed as probable based on the known CYP2C9 inhibitory mechanism of Fluconazole, clear temporal association, positive dechallenge, and absence of alternative causative factors. The events are assessed as expected based on current Fluconazole reference safety information.

CASE 2: Pediatric Hepatotoxicity - Ceftriaxone

Source Case Description

A 7-year-old male (22kg) with bacterial meningitis was prescribed Ceftriaxone 100mg/kg/day IV. On day 3 of treatment, the child developed jaundice, dark urine, and abdominal pain. LFTs showed ALT 380 U/L (normal <45), AST 290 U/L (normal <40), and total bilirubin 4.2 mg/dL. Ceftriaxone was discontinued and supportive care initiated. LFTs normalized over 3 weeks. Parents reported the case. No prior liver disease or concurrent medications.

Section 1 - Case Validity

Assessment: VALID - All four ICH E2A minimum criteria met:

- Identifiable patient: 7-year-old male, 22kg
- Identifiable reporter: Parents (non-medically qualified - flagged as such)
- Suspect drug: Ceftriaxone 100mg/kg/day IV
- Adverse event: Drug-induced liver injury (jaundice, elevated LFTs, dark urine)

Section 2 - Patient & Drug Information

Patient: 7-year-old male, 22kg

Indication: Bacterial meningitis

Suspect Drug: Ceftriaxone 100mg/kg/day IV

Concomitant Medications: None

Action Taken: Ceftriaxone permanently discontinued; supportive care initiated

Outcome: Recovered - LFTs normalized over approximately 3 weeks

Section 3 - MedDRA Coding

AE: Jaundice

PT: Jaundice, HLT: Hepatic and hepatobiliary disorders, HLGT: Hepatobiliary disorders, SOC: Hepatobiliary disorders

AE: Dark urine

PT: Urine discolouration, HLT: Urinary tract signs and symptoms, HLGT: Urinary tract disorders, SOC: Renal and urinary disorders

AE: Abdominal pain

PT: Abdominal pain, HLT: Gastrointestinal signs and symptoms, HLGT: Gastrointestinal and abdominal conditions NEC, SOC: Gastrointestinal disorders

AE: High ALT

PT: Alanine aminotransferase increased, HLT: Liver function analyses, HLGT: Liver function analyses, SOC: Investigations

AE: High AST

PT: Aspartate aminotransferase increased, HLT: Liver function analyses, HLGT: Liver function analyses, SOC: Investigations

AE: High bilirubin

PT: Blood bilirubin increased, HLT: Liver function analyses, HLGT: Liver function analyses, SOC: Investigations

Section 4 - Seriousness Assessment

Assessment: SERIOUS

Criteria met: Medically significant - ALT 8x ULN, AST 7x ULN meeting Hy's Law threshold for severe drug-induced liver injury; required hospitalization and supportive care. Pediatric population warrants heightened seriousness consideration.

Section 5 - Causality Assessment**Assessment: PROBABLE (WHO-UMC)**

- Clear temporal relationship - onset day 3 of Ceftriaxone therapy
- Positive dechallenge - LFTs normalized over 3 weeks following discontinuation
- No concurrent medications; no prior hepatic disease
- Ceftriaxone-induced hepatotoxicity is documented in literature, particularly in pediatric patients
- No rechallenge performed
- Reporter note: Non-medically qualified reporter (parents) - medical assessment based on clinical data provided

Section 6 - Expectedness Assessment**Assessment: EXPECTED (Labeled)**

Ceftriaxone prescribing information lists hepatic enzyme elevations and hepatotoxicity as known adverse events. Event is consistent with reference safety information.

Section 7 - Case Narrative

This is a spontaneous report from the parents of a 7-year-old male patient (22kg) with a known diagnosis of bacterial meningitis and no prior history of hepatic disease or concurrent medications.

The patient was initiated on Ceftriaxone 100mg/kg/day administered intravenously for treatment of bacterial meningitis. On day 3 of treatment, the patient developed jaundice, dark urine, and abdominal pain. Laboratory investigations revealed significantly elevated liver function tests with ALT 380 U/L (reference <45 U/L), AST 290 U/L (reference <40 U/L), and total bilirubin 4.2 mg/dL, consistent with drug-induced liver injury.

Ceftriaxone was discontinued and supportive care was initiated. Liver function tests normalized over a period of approximately 3 weeks. No rechallenge was performed. The case was reported by the patient's parents (non-medically qualified reporter).

This case is assessed as serious based on medical significance, with LFT elevations exceeding 8x upper limit of normal meeting criteria for severe hepatotoxicity. Causality is assessed as probable based on clear temporal association, positive dechallenge, and absence of alternative causative factors including concurrent medications or pre-existing liver disease. The events are assessed as expected based on current Ceftriaxone reference safety information.

CASE 3: Biologic-Induced Immune-Mediated Pneumonitis - Pembrolizumab

Source Case Description

A 54-year-old female with metastatic non-small cell lung cancer was receiving Pembrolizumab 200mg IV every 3 weeks. After cycle 4, she developed progressive dry cough, dyspnea on exertion, and low grade fever. CT chest revealed bilateral ground glass opacities. Pneumonitis was diagnosed. Pembrolizumab was permanently discontinued. She was started on high dose corticosteroids. Partial improvement noted after 6 weeks but residual breathlessness persisted. The oncologist reported the case.

Section 1 - Case Validity

Assessment: VALID - All four ICH E2A minimum criteria met:

- Identifiable patient: 54-year-old female
- Identifiable reporter: Oncologist (medically qualified)
- Suspect drug: Pembrolizumab 200mg IV Q3W
- Adverse event: Immune-mediated pneumonitis

Section 2 - Patient & Drug Information

Patient: 54-year-old female

Indication: Metastatic non-small cell lung cancer (mNSCLC)

Suspect Drug: Pembrolizumab 200mg IV every 3 weeks

Cycle at Onset: After cycle 4

Action Taken: Permanently discontinued; high dose corticosteroids initiated

Outcome: Partially recovered - residual breathlessness persisting at time of report. Follow-up pending.

Section 3 - MedDRA Coding

AE: Dry cough

PT: Cough, HLT: Breathing abnormalities, HLGT: Respiratory disorders NEC, SOC: Respiratory, thoracic and mediastinal disorders

AE: Breathlessness on exertion

PT: Dyspnoea exertional, HLT: Breathing abnormalities, HLGT: Respiratory disorders NEC, SOC: Respiratory, thoracic and mediastinal disorders

AE: Low grade fever

PT: Pyrexia, HLT: Body temperature conditions, HLGT: Body temperature conditions and temperature intolerance, SOC: General disorders and administration site conditions

AE: Pneumonitis

PT: Pneumonitis, HLT: Lower respiratory tract inflammatory and immunologic conditions, HLGT: Lower respiratory tract disorders, SOC: Respiratory, thoracic and mediastinal disorders

Section 4 - Seriousness Assessment

Assessment: SERIOUS

Criteria met: Life-threatening - bilateral ground glass opacities on CT with persistent residual breathlessness after 6 weeks; required permanent discontinuation of therapy and high dose corticosteroid treatment. Medically significant immune-mediated reaction with incomplete recovery.

Section 5 - Causality Assessment

Assessment: PROBABLE (WHO-UMC)

- Temporal relationship established - onset following cycle 4 of Pembrolizumab
- Positive dechallenge - partial improvement noted after permanent discontinuation
- Immune-mediated pneumonitis is a well-characterized class effect of PD-1 inhibitors
- No alternative causative factor identified
- Oncologist clinical assessment confirms immune-mediated etiology
- Note: Incomplete recovery - residual breathlessness persisting; follow-up information requested

Section 6 - Expectedness Assessment

Assessment: EXPECTED (Labeled)

Pembrolizumab (Keytruda) prescribing information lists immune-mediated pneumonitis including fatal cases as a known serious adverse reaction with a black box warning. Event is consistent with reference safety information.

Section 7 - Case Narrative

This is a spontaneous report from an oncologist concerning a 54-year-old female patient with metastatic non-small cell lung cancer receiving Pembrolizumab 200mg intravenously every 3 weeks.

Following completion of cycle 4 of Pembrolizumab, the patient developed progressive dry cough, dyspnea on exertion, and low grade fever. CT chest imaging revealed bilateral ground glass opacities. A diagnosis of immune-mediated pneumonitis was established by the treating oncologist.

Pembrolizumab was permanently discontinued. High dose corticosteroid therapy was initiated. Partial clinical improvement was noted after 6 weeks of treatment; however, residual breathlessness persisted at the time of reporting. Follow-up information has been requested to determine final outcome.

This case is assessed as serious based on the life-threatening nature of bilateral pneumonitis requiring permanent treatment discontinuation and high dose corticosteroid therapy, with incomplete recovery at time of report. Causality is assessed as probable based on temporal association following cycle 4, positive dechallenge with partial improvement, established mechanism of immune-mediated toxicity with PD-1 inhibitors, and oncologist assessment. The events are assessed as expected based on current Pembrolizumab reference safety information.

All cases processed in accordance with ICH E2A guidelines. MedDRA coding based on current hierarchy conventions. Causality assessed using WHO-UMC categories. This portfolio was prepared as a self-initiated learning exercise to demonstrate ICSR processing competency.